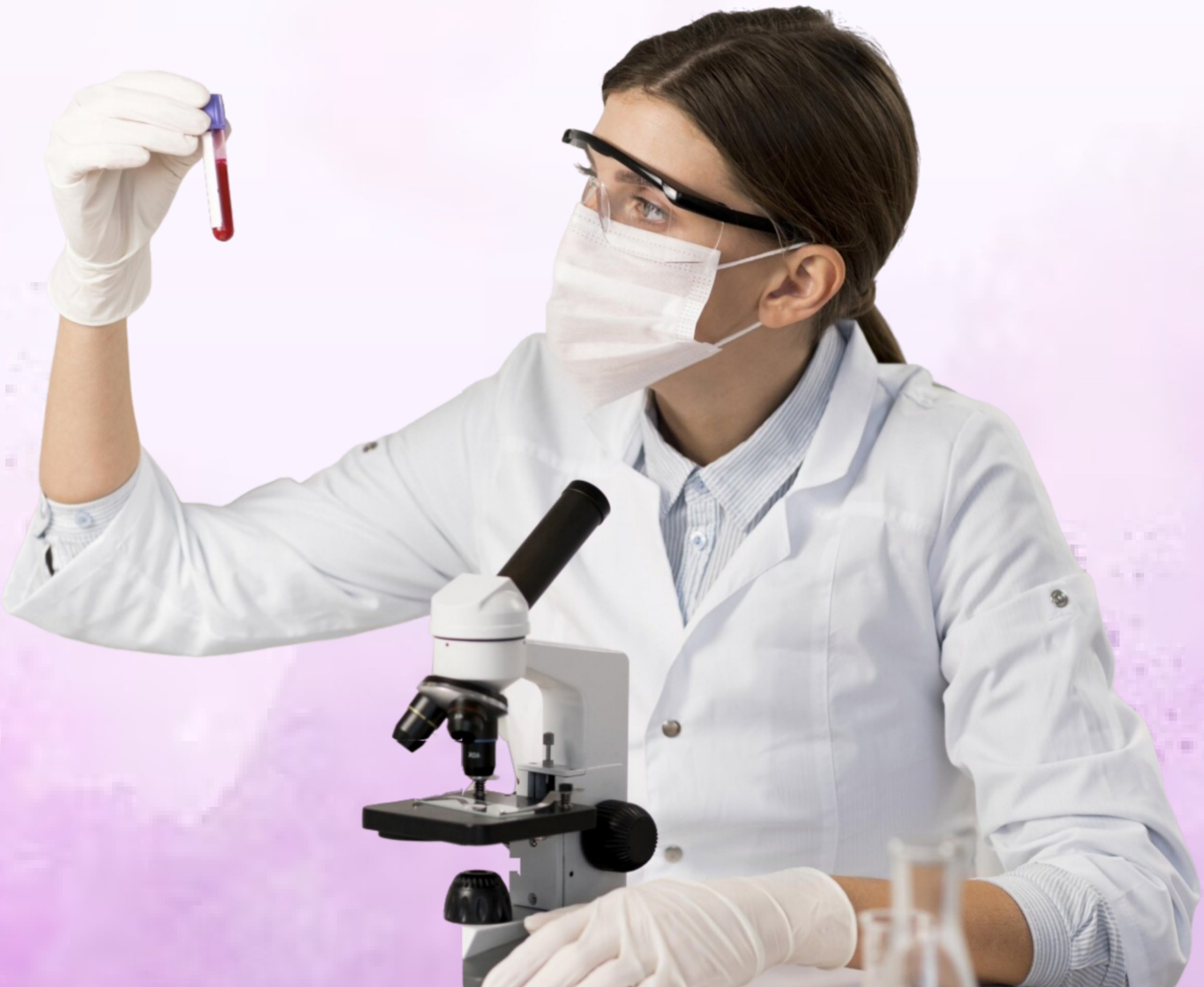
CRYOVIVA
LIFESCIENCES
THE BANK OF LIFE



UMBILICAL CORD BLOOD REPORT

DOCUMENT NO.: CLS/GM/UCB REPORT/v. 01/09/2023-01

Cryoviva Life Sciences Pvt. Ltd.
1st Floor, A Wing, Plot No. 19, Sector 35, Gurugram-122004, Haryana (India)

Dear Parents,
Thank you for choosing Cryoviva Life Sciences for the cryopreservation of your Baby's precious Stem Cells.

We have completed all the required tests for the cryopreservation of your Baby's Stem Cells at Cryoviva Life Sciences' facility and enclosed herewith are the test reports.

For your reference and understanding 'interpretation' of test reports with relevant information is provided.

We would advise you to keep this document safely filed for future use. In the event of a necessity for retrieval of the Specimen related information, please remember to quote your Client Primary ID No. to us and we will initiate the retrieval procedure.

In case of any queries and/or change in your contact information please get in touch with us at support@cryovivalifesciences.in and for availing our Refer-a-Friend program, you may call us at:

188001019587 (Toll-free)

support@cryovivalifesciences.in

Your sincerely
For Cryoviva Life Sciences Pvt. Ltd.

Lab Director	**Medical Director**

DOCUMENT NO.: CLS/GM/UCB REPORT/v. 01/09/2023-01

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Name of Client	
Primary ID	
UCB Product ID	
Plan Opted for	
Contact Number	
Date & Time of Umbilical Cord Blood Collection (hrs)	
Date & Time of Umbilical Cord Blood Processing (hrs	
Date & Time of Report Generation (hrs)	

Status of the Report	Final Report ☐	Intermediate ☐	Duplicate ☐	Amended ☐
Date of Report Release				
Reason of Amendment				
Remarks				

DOCUMENT NO.: CLS/GM/UCB REPORT/v. 01/09/2023-01

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UMBILICAL CORD BLOOD REPORT

Name of Client	
Client Primary ID No.	
Lab ID No.	
Volume of Umbilical Cord Blood Collected (ml)	
Date & Time of Umbilical Cord Blood Collection (hrs)	
Date & Time of Report Generation (hrs)	

STORAGE REPORT

Product Name	
Storage Container	
Volume of Final Product Stored (ml)	
Date & Time of Processing (hrs)	
Storage Temperature	At or below -150°C in Vapour Phase of Liquid Nitrogen Tank
Remarks	

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INTERPRETATION

Volume of Umbilical Cord Blood Collected

It denotes the amount of Umbilical Cord Blood collected from the placenta and umbilical cord vessels after the baby has been delivered and separated from mother. The collection volume varies from 10 to 120 ml or more, and is collected in a sterile collection bag containing a known volume of anticoagulant.

Volume of Umbilical Cord Blood Stored

This is the volume of Umbilical Cord Blood stored after concentration of the nucleated blood cells (including the Stem Cells). During processing of the Specimen, most of the RBCs and plasma is removed (since these are not required for transplantation), and cryoprotectant is added for making it up a standard volume of 26ml, out of which usable volume is 25ml and approx. 1ml is used for testing.

Nature of Packing

The minimally-manipulated product, consisting of nucleated blood cells including Stem Cells in autologous plasma mixed with cryoprotectant is stored in a labeled Cryogenic Storage Bag individually sealed in overwrap bag and metal cassette. The metal cassette serves to prevent Specimen mix up, minimize thermal stress and provide physical protection.

Storage Condition

The metal cassette containing Umbilical Cord Blood Specimen Specimen is stored in cryogenic condition in Liquid Nitrogen Storage Tank, until it is retrieved for use in transplant.

Every 25 ml Umbilical Cord Blood Unit is stored continuously below minus 150°C in the vapor phase of liquid nitrogen. This temperature ensures maximum viability of the Stem Cells for extended period of time.

Remarks	

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Name of Client	
Client Primary ID No.	
Lab ID No.	
Volume of Umbilical Cord Blood Collected (ml)	
Date & Time of Umbilical Cord Blood Collection (hrs)	
Date & Time of Report Generation (hrs)	

SPECIMEN TYPE: Umbilical Cord Blood

Parameter	Method	Result	Unit	Reference Range (Expected Result)
Total Viable Nucleated Cell Count (CD45+)	Flow Cytometry Instrument- BD FACS Canto II Software- BD FACS Canto		10^8	
Total Viable Hematopoietic Stem Cell Count (CD34+)	Flow Cytometry Instrument- BD FACS Canto II Software- BD FACS Canto		10^6	
Nucleated RBC Count	Leishman's Stain		Per 100 WBC	
Umbilical Cord Blood Group, ABO & Rh Typing	Haemagglutination ABO Rh Neonate Gel Card Tube Method (Forward Grouping)			

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INTERPRETATION

Total Viable Nucleated Cell Count (CD45+)

The Total Viable Nucleated Cell Count refers to the actual number of live nucleated cells; white blood corpuscles (WBC) and haematopoietic stem cells (CD34+ cells) present in the processed Umbilical Cord Blood Specimens. For your and the transplant Physician's help, Cryoviva reports the actual number of live nucleated cells (measured by flow cytometry), as this is one of the primary indicators for the efficacy of successful transplantation.

Total Hematopoietic Cell Count (CD34+)

CD34 is a hematopoietic progenitor cell marker. This is the total number of hematopoietic Stem Cells present in the processed Umbilical Cord Blood product after processing and is measured by flow cytometry.

Nucleated RBC Count

Nucleated Red Cell Count is the number of nucleated RBC (Red Blood Cells) present in the processed Umbilical Cord Blood product after processing per 100 nucleated cell count.

Umbilical Cord Blood Group

This refers to the ABO blood grouping & Rh typing of the collected Umbilical Cord Blood. However, it should be noted that the Anti A and Anti B antibodies are not fully developed at birth, but develops later and remains throughout life. Anti A and Anti B produced by an infant can generally be detected in serum after 3 to 6 months of life. Hence it is advised to determine the blood group from the baby's blood directly in future from a diagnostic laboratory.

Remarks	

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UMBILICAL CORD BLOOD REPORT

Name of Client	
Client Primary ID No.	
Lab ID No.	
Volume of Umbilical Cord Blood Collected (ml)	
Date & Time of Umbilical Cord Blood Collection (hrs)	
Date & Time of Report Generation (hrs)	

SPECIMEN TYPE: Umbilical Cord Blood

Parameter	Method	Result	Unit	Reference Range (Expected Result)
Aerobic Bacterial/ Fungal Culture				
Screening Result	BD BACTEC			
*Final Result				
Anaerobic Bacterial Culture				
Screening Result	BD BACTEC			
*Final Result				

*Tests performed as per requirement when the screening assays are Positive for the respective parameters.

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INTERPRETATION

Sterility

The Screening test is performed to check presence or absence of Aerobic and Anaerobic microbial contamination on every Umbilical Cord Blood Specimen by using Automated Blood Culture Instrument. If the screening result is flagged as "Positive", the Specimen is sent to an Accredited Reference Laboratory for further microbial culture and sensitivity to get the final result.

Storage Decision

"Microbial Culture and Antibiotic Sensitivity" are performed only for the Microbial screening positive Specimens to identify the presence of Aerobic Bacterial/Fungal and Anaerobic Bacterial/Fungal contamination.

A positive Microbial culture does not necessarily denote infection of either the mother and/or the new born, it may just denote, carriage of microbes from the surface of the cord during collection by the needle tip flushed into the Umbilical Cord Blood thus collected. It also does not render the Specimen unfit for storage or use in future treatments, such Specimens are still good for transplantation, final decision will depend on the transplant physician.

Remarks	

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MATERNAL BLOOD REPORT

Name of Client	
Client PID	
Lab ID No.	
Volume of Maternal Blood Specimen Collected (ml)	
Date & Time of Maternal Blood Collection (hrs)	
Date & Time of Report Generation (hrs)	

SPECIMEN TYPE: Maternal Blood (Collected in K_2EDTA & SST)

Parameter	Method	Result	Unit	Reference Range (Expected Result)
1. Hepatitis B Virus (HBV) - Hepatitis B Surface antigen - HBV DNA	CMIA NAAT			
2. Hepatitis C Virus (HCV) - Anti-HCV antibody - HCV RNA	CMIA NAAT			
3. Human Immunodeficiency Virus (HIV 1 & 2) - Anti HIV 1 & 2 & p24 antigen - HIV 1 (Subtypes M & O) & HIV 2 RNA	CMIA NAAT			
4. Cytomegalovirus (CMV) - CMV IgM - CMV IgG	CMIA CMIA			
5. Human T-Lymphotropic Virus (HTLV I/II) -*HTLV I/II Antibody Screening	EIA			
6. Treponema pallidum (Syphilis) - *Syphilis Screening	RPR			
7. Malaria Antigen	RIA			

*Tests performed as per requirement when the screening assays Reactive for the respective parameters

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INTERPRETATION

HEPATITIS B

Hepatitis B surface Antigen: Hepatitis B surface antigen is the diagnostic marker for screening of Hepatitis B infection which causes inflammation of liver. It can be detected in high levels in serum during acute or chronic hepatitis B virus infection. This antigen is the earliest indicator of acute hepatitis B and frequently identifies infected people before symptoms appear Perinatal Hepatitis B virus infection may be possible due to transmission from infected mother to child. Hepatitis B a blood borne infection, can be transmitted by exposure of blood/body fluids through infected needles, instruments, medical/dental procedures, acupuncture or tattoos performed with poorly sterilized equipments, blood transfusions and sexual activity.

HBV DNA: Hepatitis B virus (HBV) DNA is tested by Nucleic Acid Amplification Test (NAAT), the most specific & sensitive test to identify presence of Hepatitis -B virus in the Specimen. This test can identify the presence of Hepatitis-B virus during the "Window period" and long before it could be identified in serological tests.

Storage decision: If the maternal blood is screened to be HBV positive, then it may not be usable for transplant, except if the transplant physician approves for autologous use only. The Client will be intimated accordingly.

HEPATITIS C VIRUS (HCV)

Anti-HCV Antibody: Anti-HCV test looks for antibodies of the Hepatitis C virus in blood. Hepatitis C Virus (HCV) is a hepatropic virus which is one of the major causes of liver disease and a potential cause of substantial morbidity and mortality worldwide. Hepatitis C can be transmitted by exposure of blood/body fluids through infected needles, instruments, medical/dental procedures, acupuncture or tattoos performed with poorly sterilized equipments and sexual activity. The virus, estimated to infect about 3% of the world population, is primarily transmitted via the parental route which includes injection drug use, blood transfusion, unsafe injection practices and other healthcare related procedures.

HCV RNA:

Hepatitis C virus (HCV) RNA is tested by Nucleic Acid Amplification Test (NAAT), the most specific & sensitive test to identify presence of Hepatitis-C virus in the Specimen. This test can identify the presence of Hepatitis-C virus during the "Window period" and long before it could be identified in serological tests.

Storage decision: If the maternal blood is screened to be HCV positive, then it may not be usable for transplant, except if the transplant physician approves for autologous use only. The Client will be intimated accordingly.

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HUMAN IMMUNODEFICIENCY VIRUS (HIV 1 & 2)

Human Immunodeficiency Virus HIV-1 or HIV-2 are retroviruses, preferentially infecting CD4+ T lymphocytes in lymph nodes and other lymphoid tissue. HIV a blood borne virus, can be transmitted from individuals through body fluids, primarily blood and semen. Intravenous drug users and persons receiving blood transfusions can be exposed to the virus through infected blood or body products. Sexual transmission is mainly the result of the transfer of and exposure to infected semen. Women as well as men can transmit the virus sexually. A baby may become infected during pregnancy, delivery, or when breast feeding if its mother has the infection. A person may carry the virus for months before testing positive and may carry the virus for months or years before the symptoms appear. An HIV positive person can still spread the disease even though he or she may appear healthy. When HIV enters the blood stream it invades and destroys cells in the body's infection and cancer fighting system and reduces the body's ability to fight infections. HIV infection leads to AIDS when the infected person's immune system is compromised to an extent that allows the onset of opportunistic infections.

THE HIV TEST AND VOLUNTARY TESTING

The HIV tests are blood tests for the presence of HIV infection or exposure to the virus. A positive test result means that you have been exposed to the virus. It may not mean that you have AIDS now or that you will become sick with AIDS in the future. A negative test means that you are probably not infected with the virus. Limitation of the test is the “window period” which is the time of infection to time of detection.

Anti HIV1 & 2, p24 Antigen: Anti-HIV test detects antibodies against HIV-1 & 2. HIV p24 antigen is done for the presence of HIV virus particle. HIV-1 & 2 RNA: Human Immunodeficiency (HIV) RNA is tested by Nucleic Acid Amplification Test (NAAT), the most specific & sensitive test to identify presence of Human Immunodeficiency Virus 1 & 2 virus in the maternal blood Specimen. This test can identify the presence of HIV during the “Window period” and long before it could be identified in serological tests.

Storage decision: If the maternal blood is screened to be HIV-1 & 2 positive, then Cryoviva will not store the Umbilical Cord unit and discard the same. The Client will be intimated accordingly.

Note: As the HIV tests performed on your blood are Screening tests, we advise you to get medical consultation, counseling and testing at the nearest ICTC (Integrated Counselling and Testing Centre), in any Government Hospital for confirmation of status and further management. We would be pleased to provide you with any clarifications you require in this regard.

CYTOMEGALOVIRUS (CMV)

Cytomegalovirus IgM: Cytomegalovirus (CMV) is a member of the Herpesviridae family of viruses and usually causes asymptomatic infection, primarily within bone marrow derived cells. CMV infection in immunocompetent individuals may manifest as a mononucleosis-type syndrome, similar to primary Epstein-Barr virus infection, with fever, malaise and lymphadenopathy. CMV IgM positive results indicate a recent infection (primary, reactivation or reinfection). CMV IgM antibody response in secondary CMV infections have been demonstrated in some CMV mononucleosis patients, in a few pregnant women and in renal and cardiac transplant patients. Levels of antibodies may be lower in transplant patients with secondary rather than primary infections.

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Cytomegalovirus IgG:
CMV IgG is the antibody to CMV. A majority of adults have IgG antibodies consistent with past infection. CMV IgG reactive indicates acquired immunity towards CMV.

Storage decision:
In the event the maternal blood is found in house positive for CMV IgM, then the Specimen may not be usable for Transplant, unless approved by Transplant physician for autologous use only. The client shall be intimated accordingly.

HUMAN T-CELL LYMPHOTROPIC VIRUS (HTLV I/II)

Human T-cell Lymphotropic Virus I/II: The Human T cell Lymphotropic Virus type I and II (HTLV I and HTLV II) infect the white blood cells (T4 lymphocytes). HTLV I infection has been shown to be associated with T and B cell Chronic Lymphocytic Leukaemia (CLL), multiple myeloma, arthritis, Kaposi's sarcoma, and some cases of non-Hodgkin's lymphoma. Both HTLV I/II can be transmitted by sexual contact, transfusion of cellular blood products, intravenous drug injection using infected needles, and from mother to child due to breast feeding. HTLV I/II screening reactive determines the presence of antibodies towards both HTLV I and HTLV II (HTLV I/II Antibody). The positive result may affect the storage/use of Umbilical Cord unit.

SYPHILIS

Syphilis is a sexually transmitted disease, caused by Treponema pallidum. It can cause serious health problems if it is not treated properly. Syphilis is divided into stages (primary, secondary, latent and tertiary). It can be spread by direct contact with syphilis sore during vaginal, anal or oral sex. It can spread from an infected mother to her unborn baby through vertical transmission. Test of syphilis through RPR is a screening assay and a reactive result is suspicious of active disease which needs to be tested by further assays.

MALARIA

Malaria is caused by protozoa, 4 subtypes of which are known to exist and pathogenic to human beings Plasmodium falciparum, P. vivax, P. malariae and P. ovale. They are generally transmitted through bites of infected Anopheles mosquitoes, but can also be transmitted via transfusion of infested blood.

Storage decision: In the event the maternal blood is found in house positive for Malaria, then Cryoviva will discard the Umbilical Cord Unit and the client will be intimated accordingly.

Remarks	

Lab Director **Medical Director**

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PROCESS FLOW OF YOUR SAMPLE

1 Your Specimen

2 Received at Lab on

3 UCB Processing Completed on

4 Report Released on

YOUR PRECIOUS SPECIMEN IS IN SAFE CONDITION

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CONDITIONS OF LABORATORY TESTING & REPORTING

1. All the test has been performed to check the quality of the product for cryogenic storage of the Specimen in **CLSPL Laboratory**. It has no relation with the clinical condition of the client.

2. **CLSPL** confirms that all tests have been performed or assayed with highest quality standards clinical safety and technical integrity.

3. For any query regarding the test results, the report can be shared or discussed with the consultant physician.

4. Tests performed for Transfusion Transmissible Disease (T.T.D) parameters are screening in nature, not be considered as confirmed report.

5. In case of queries or unexpected test result please call at **188001019587**

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STORAGE CERTIFICATE

This is to certify that

UMBILICAL CORD BLOOD STEM CELL PRODUCT

has been successfully cryopreserved at **Cryoviva Life Sciences**
on ....

Lab Director

Medical Director

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